

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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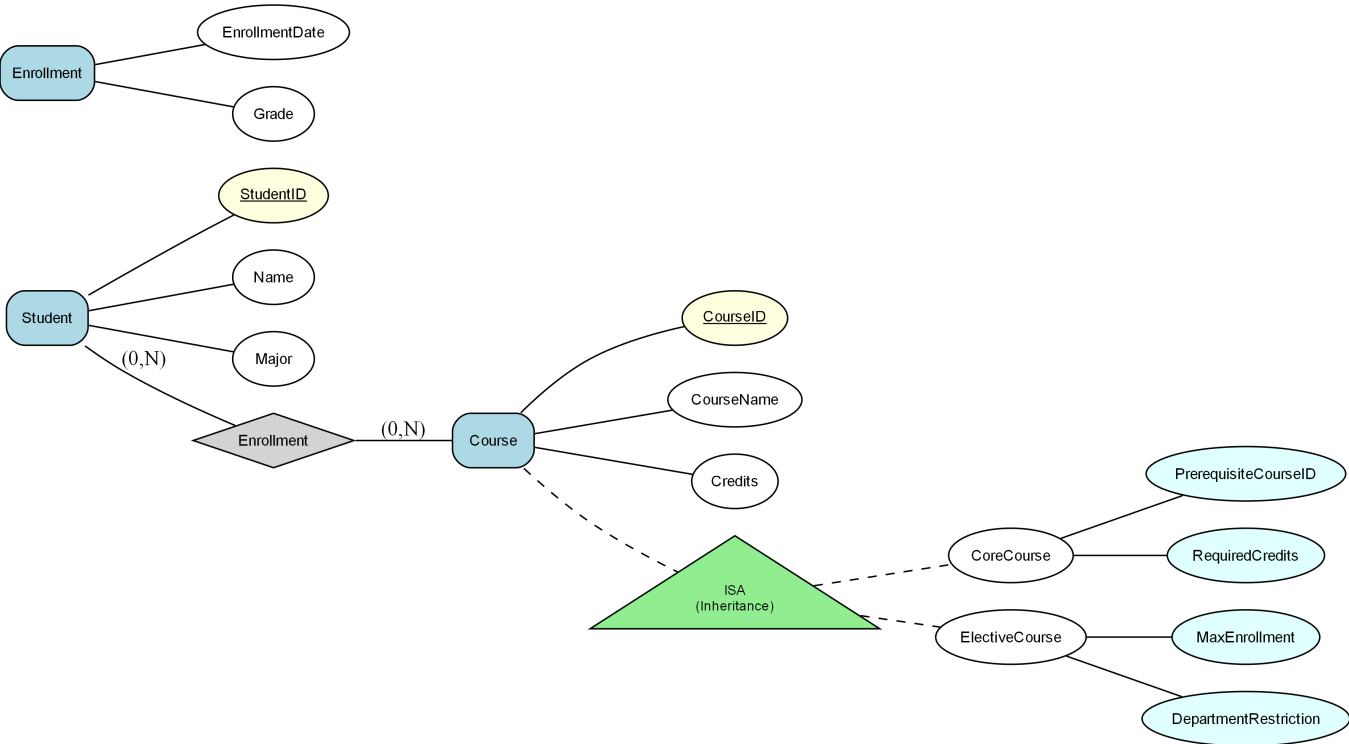
Total Marks: 100

Examination Paper

Question Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3)

Question Q2 (15 marks)

Consider a relation schema: Students (StudentID, StudentName, CourseID, AdvisorID) with the functional dependencies $F = \{ \text{StudentID} \rightarrow \text{StudentName}, \text{StudentID} \rightarrow \text{CourseID}, \text{CourseID} \rightarrow \text{AdvisorID} \}$.

a) Draw the functional dependency diagram. (1)

b) Use the attribute closure and identify the primary key for the given relation. (1)

c) If we insert a row $\langle 123, \text{John Doe}, \text{CSE101}, \text{null} \rangle$ into the given relation, what is the

update anomaly that is violated? Give reasons. (2)

d) Assume that the relation has a row, < 456, Jane Smith, CSE102, 7>. If we remove the student Jane Smith, what is the possible update anomaly that can be violated? Give reasons. (2)

e) Using the above functional dependencies, design a set of 3NF relations for the above given relation. Show clearly each stage in deriving the 3NF relations. Also identify the normal form at each of these stages. (3)

f) Do the update anomalies of part (c) and (d) hold in the 3NF relations? Discuss briefly. (2)

g) Are the 3NF relations derived in part (e) still in 3NF? Justify your answer. (2)

h) Are the 3NF relations derived in part (e) in BCNF? If not, bring the relations to BCNF. (2)

Question Q3 (25 marks)

A retail company is implementing a new database system to manage its products, sales, and customer relationships. The database schema will include the tables: Products (product_id, product_name, category, price), Sales (sale_id, product_id, quantity, sale_date), and Customers (customer_id, customer_name, contact_info). The company aims to optimize its inventory management and improve customer service.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2)

b) Code Segment:

```
```sql
PreparedStatement pstmt = connection.prepareStatement(
 "SELECT * FROM Employees WHERE Department = ?");
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2)

c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2)

d) Briefly explain how authentication and authorization are achieved in SQL Server. (3)

e) The retail company has designated Sarah as the senior database administrator responsible for supervising the system. Write a T-SQL statement to create a login for

- Sarah with Windows authentication. (2)
- f) Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3)
- g) Assuming Emily's username is 'emily.ent', write T-SQL statements to assign her the responsibility of managing the Products table using a user-defined server role. (5)
- h) Assuming Nathan's login name is 'nathan.dev', write T-SQL statements to grant him permission to create tables and views in the Sales database. (3)
- i) Assuming Michael's login name is 'michael.ent', write T-SQL statements to allow him to perform data entry tasks on the Customers and Products databases. (3)

Question Q4 (40 marks)

Consider a database management system scenario.

- a) Define the key concepts and their importance related to RELATIONAL_ALGEBRA. (4)
- b) Describe the main components and how they interact related to RELATIONAL_ALGEBRA. (4)
- c) Get the details of flights that are scheduled on both dates 01/12/2024 and 02/12/2024 at 16:00 hours. (6)
- d) Compare different implementation strategies and their trade-offs related to RELATIONAL_ALGEBRA. (6)
- e) How many passengers have booked the flight f001? (4)
- f) Express the above queries (i) to (iv) in tuple calculus. (16)